

The Role of Whole Grains in Body Weight Regulation

Whole grain (WG)-rich diets are purported to have a variety of health benefits, including a favorable role in body weight regulation. Current dietary recommendations advocate substituting WG for refined grains (RG), because many of the beneficial bioactive components intrinsic to WG are lost during the refining process. Epidemiological studies consistently demonstrate that higher intakes of WG, but not RG, are associated with lower BMI and/or reduced risk of obesity. However, recent clinical trials have failed to support a role for WG in promoting weight loss or maintenance. Though the biochemical and structural characteristics of WG have been shown to modulate appetite, nutrient availability, and energy utilization, the capacity of WG foods to elicit these effects varies with the type and amount of grain consumed as well as the nature of its consumption. As such, WG foods differentially affect physiologic factors influencing body weight with the common practice of processing and reconstituting WG ingredients during food production likely mitigating the capacity for WG to benefit body weight regulation.